

Comparative Study of Machine Learning Models for Predicting Diabetes Hospital Readmission

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1. Abstract

Hospital readmission is a major challenge in healthcare systems, as frequent readmissions increase medical costs and may indicate inadequate treatment or follow-up care. This project investigates the use of machine learning models to predict hospital readmission among diabetes patients using the Diabetes 130 U.S. Hospitals dataset. The dataset contains structured healthcare records and presents several challenges, including missing values, categorical variables, high-cardinality diagnosis codes, and class imbalance.

It is important to observe hospital readmissions within 30 days since they are a quality metric in the healthcare system, as they are often associated with an increase in cost and potential gaps in patient care. Evaluating features such as the number of inpatient visits, discharges, and the number of diagnoses are highly relevant. These factors are expected to play a significant role in predicting readmission risk.

To address these issues, preprocessing techniques such as missing value handling, duplicate removal, one-hot encoding, feature scaling, and class weighting were applied. Four supervised learning models were implemented and compared: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting.

The models were evaluated using accuracy, precision, recall, F1-score, runtime analysis, ROC curves, and confusion matrices. Because the dataset is imbalanced, recall and F1-score were emphasized over accuracy when comparing model performance.

The results show that while some models achieved high accuracy, they were less effective at identifying patients who were actually readmitted. After addressing class imbalance, Gradient Boosting achieved the best overall performance, with the highest recall and F1-score, while Logistic Regression provided the fastest training time. These findings demonstrate the importance of selecting appropriate evaluation metrics and imbalance-handling techniques when applying machine learning to healthcare prediction tasks.

2. Introduction

Hospital readmission prediction is an important problem in healthcare analytics, as reducing readmission rates can improve patient outcomes and reduce medical costs. Diabetes is a chronic condition that often requires continuous monitoring, making it a relevant focus for predictive modeling.

The dataset used in this project consists of patient records from 130 U.S. hospitals between 1999 and 2008. It includes features such as patient demographics, medical diagnoses, laboratory results, and hospital stay information. However, the dataset presents several challenges, including missing values, categorical variables, and potential class imbalance.

The objective of this project is to build and evaluate multiple machine learning models to predict whether a patient will be readmitted to the hospital. By comparing different models, we aim to determine which approach performs best for this classification task and provides the most reliable predictions.

3. Related Work

Due to the importance of improving patient results and cutting expenses, hospital readmission prediction has been extensively researched in healthcare analytics. Many machine learning techniques, such as logistic regression, decision trees, and ensemble methods like random forests and gradient boosting, have been used in previous studies.

Research suggests that when working with structured healthcare data, ensemble models often perform better than simpler models because they are more adaptable and capable of capturing complex relationships between variables and features. In addition, recent research has investigated the use of deep learning methods, like neural networks, for readmission prediction; however, these models may hinder interpretability and typically require larger datasets to reduce overfitting.

Our approach aligns with existing research by comparing multiple supervised learning models and highlighting the significance of evaluation metrics beyond accuracy. Specifically, our focus

on recall and F1-score aligns with previous research recognizing the importance of correctly identifying high-risk patients in an unbalanced healthcare dataset.

4. Methodology

4.1. Data Processing

To ensure the dataset is suitable for model training, the following preprocessing steps will be applied.

First, **missing values** will be handled carefully. The dataset contains entries such as “?” or unknown values, which can negatively impact model performance. These will either be removed or imputed depending on their importance. For categorical variables, missing values will be replaced using the mode, while for numerical variables, statistical methods such as mean or median imputation will be used. This approach preserves as much data as possible while minimizing bias.

Second, **categorical variables** (such as race, gender, and diagnosis codes) will be transformed using **one-hot encoding**. This is necessary because most machine learning models require numerical input. One-hot encoding was chosen over other methods to ensure that categorical data is represented without introducing ordinal relationships where none exist.

Third, **data cleaning** will be performed by removing duplicate records and filtering out irrelevant or misleading entries. For example, patients who died during hospitalization or were discharged to hospice care may not be appropriate for readmission prediction, since their outcomes differ from typical cases. Removing these ensures the model focuses on meaningful patterns.

Finally, we will examine **class imbalance**, as readmitted patients may represent a smaller portion of the dataset. If imbalance is significant, techniques such as resampling (oversampling or undersampling) may be applied. This is important because imbalanced data can cause models to favor the majority class and reduce predictive performance for readmissions. Additionally, the dataset was split into training and testing sets using an 80/20 split with a fixed random state to ensure reproducibility.

4.2. Model Selection

To improve model performance, hyperparameter tuning was performed using grid search and cross-validation for selected models. This ensured a fair comparison and reduced the likelihood of overfitting.

Logistic Regression will be used as a baseline model because it is simple, interpretable, and widely used for binary classification problems. It provides a reference point to evaluate whether more complex models offer meaningful improvements.

Decision Trees will be included because they are highly interpretable and can capture non-linear relationships in the data. To further improve the performance of the model, hyperparameter tuning was applied using GridSearchCV with cross validation. The most impactful parameters to be optimized were the maximum tree depth, minimum samples per split, and minimum samples per leaf. To deal with the class imbalance of the dataset and to further improve the model's ability to detect readmission, class weighting was integrated. In a healthcare context, interpretability is important for understanding which factors influence readmission risk.

Random Forests will be used to improve upon decision trees by reducing overfitting through ensemble learning. By combining multiple trees, random forests provide more stable and accurate predictions, especially when dealing with complex datasets with complex features.

Gradient-boosted Trees will also be implemented because they are known for achieving high performance in structured data tasks. These models iteratively correct errors from previous trees, making them effective at capturing complex patterns in the dataset.

Using multiple models allows us to **compare the trade-offs between interpretability and performance**, which is especially important in healthcare applications where both accuracy and explainability matter.

4.3. Evaluation Metrics

To evaluate model performance, we will use several classification metrics that provide a comprehensive assessment:

Accuracy will measure the overall correctness of predictions. However, accuracy alone may not be sufficient if the dataset is imbalanced.

Precision will measure how many predicted readmissions are actually correct. This is important to avoid falsely labeling patients as high-risk.

Recall will measure the model's ability to correctly identify patients who are truly readmitted. This is especially critical in healthcare, as missing a high-risk patient could lead to serious consequences.

F1-Score will be used to balance precision and recall, providing a more reliable metric when dealing with imbalanced data.

In particular, these metrics are emphasized due to the imbalanced nature of the dataset. As we observed in our results, models resulting in higher accuracy can still perform poorly when identifying readmitted patients, making recall and F1-score more reliable factors of performance in this project.

5. Results

5.1. Evaluation Metrics & Visualizations

The four models were evaluated using accuracy, precision, recall, F1-score, training time, and prediction time. Since the goal of this project is to predict whether a patient will be readmitted within 30 days, the positive class is “readmitted < 30” and the negative class is “not < 30.” Recall and F1-score are especially important because they measure how well each model identifies patients who are actually readmitted within 30 days.

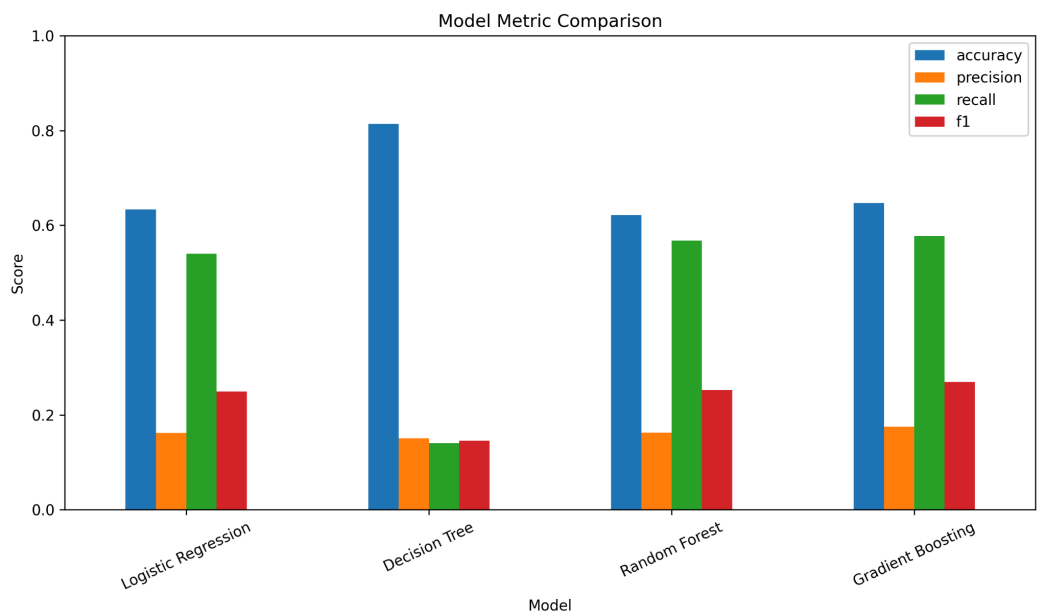
Model	Accuracy	Precision	Recall	F1-Score	Training Time (s)	Prediction Time (s)
Logistic Regression	0.6336	0.1621	0.5403	0.2494	5.29	0.04
Decision Tree	0.8139	0.1506	0.1405	0.1454	59.54	0.10
Random Forest	0.6220	0.1626	0.5674	0.2528	11.23	0.10
Gradient Boosting	0.6470	0.1755	0.5771	0.2692	108.06	0.16

Fig. 1.

Comparison of performance metrics across all models

While the baseline Decision Tree model achieved higher accuracy, its recall and F1 score were relatively low, which indicates poor performance on the minority class as seen in the above table. However, after applying hyperparameter tuning, the model showed significant improvement in both recall and F1-score with the values of approximately 0.54 and 0.26 respectively. As a result, the improvement is a demonstration of how tuning can impact the performance of the model, especially on a dataset as imbalanced as this one.

This improvement is also reflected in the confusion matrix (Fig. 3) and precision-recall curve (Fig. 9) where the tuned Decision Tree shows higher sensitivity to the minority class while maintaining a balance between precision and recall.



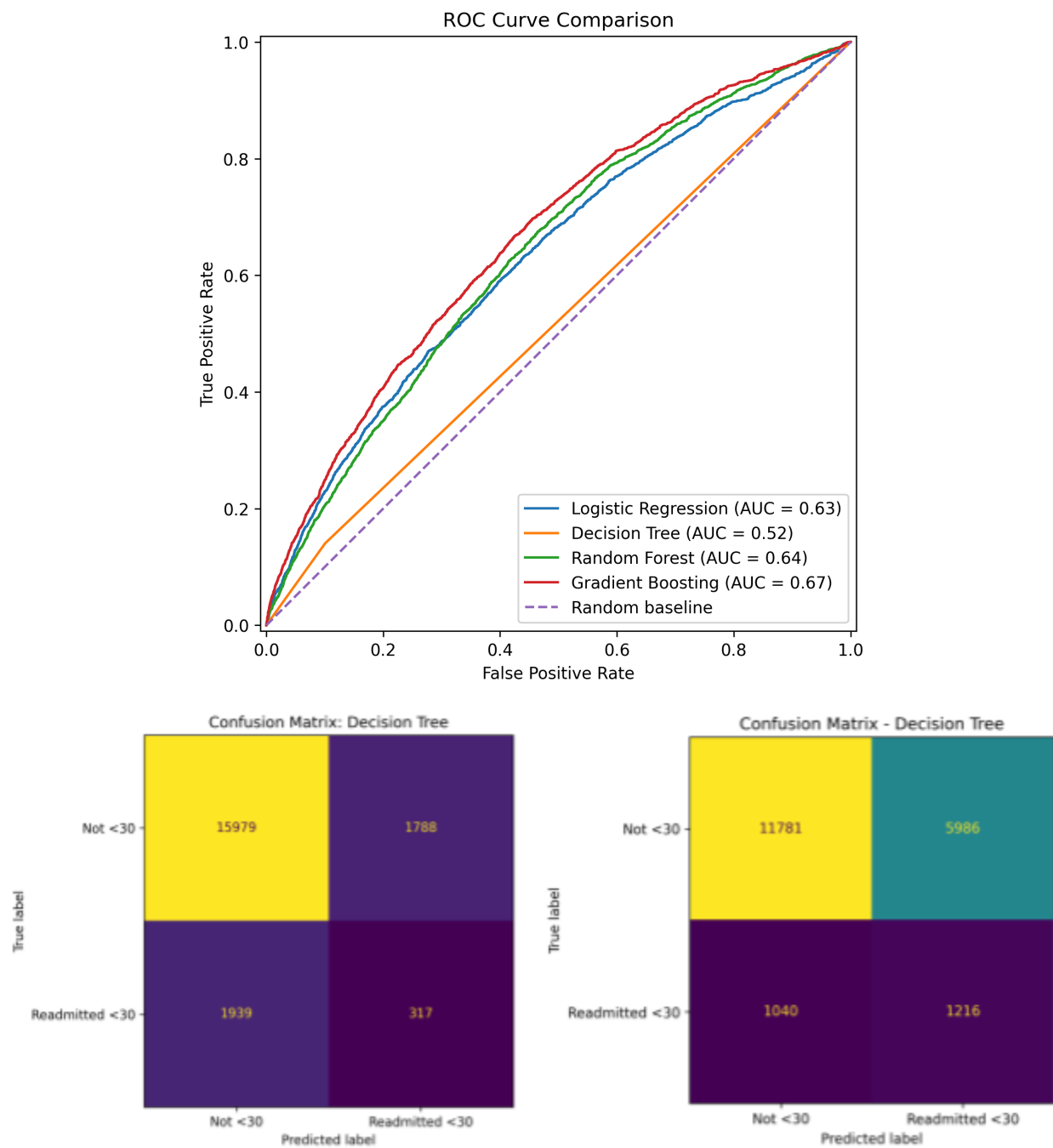
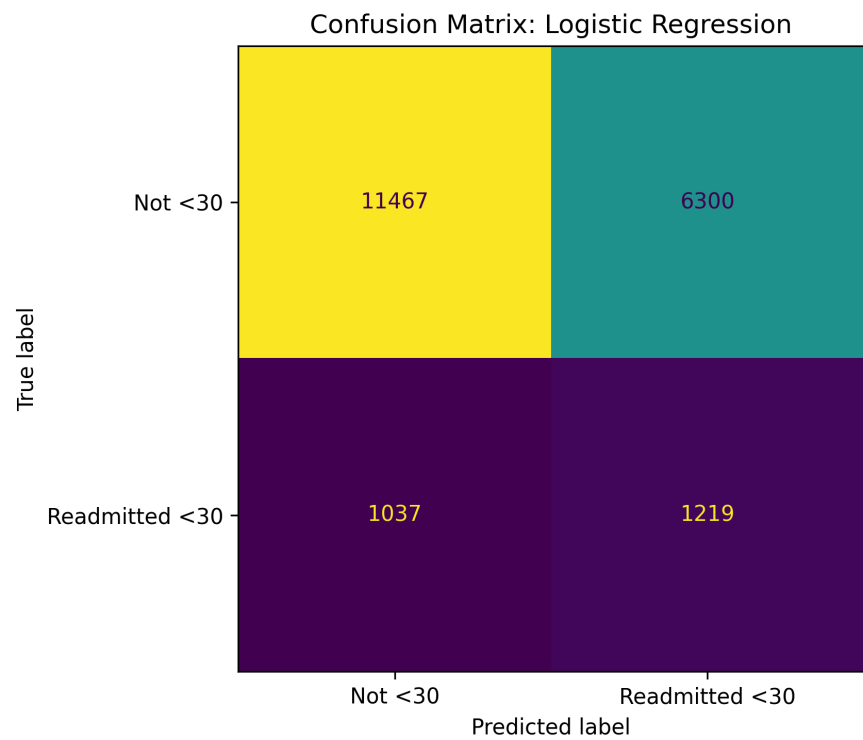
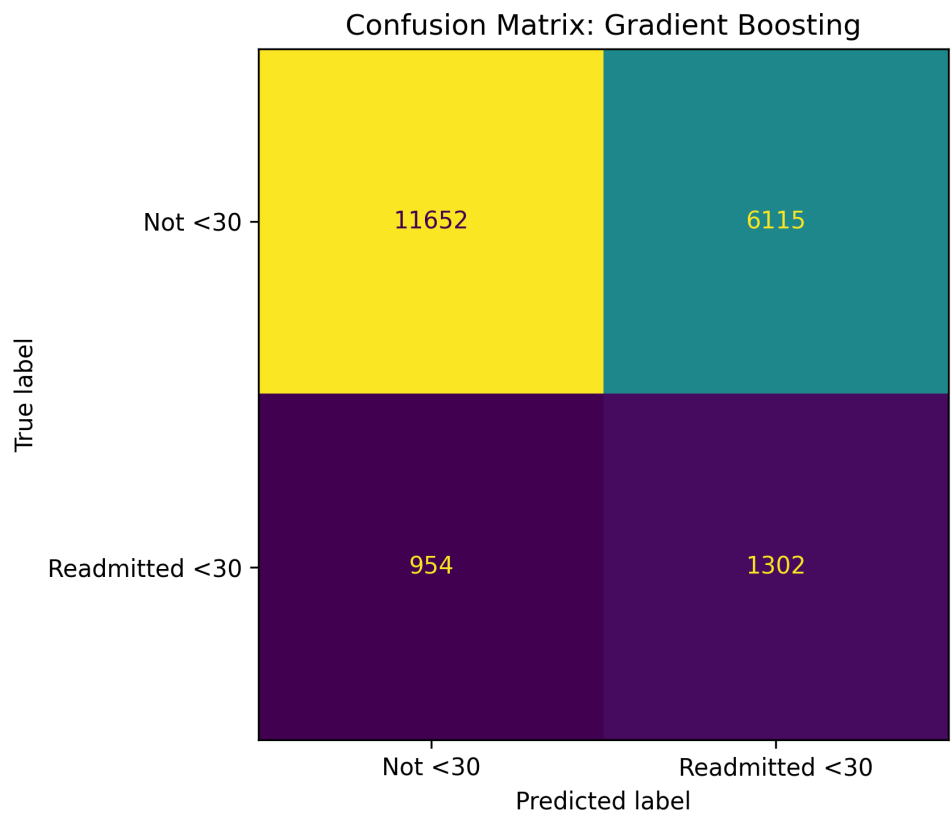
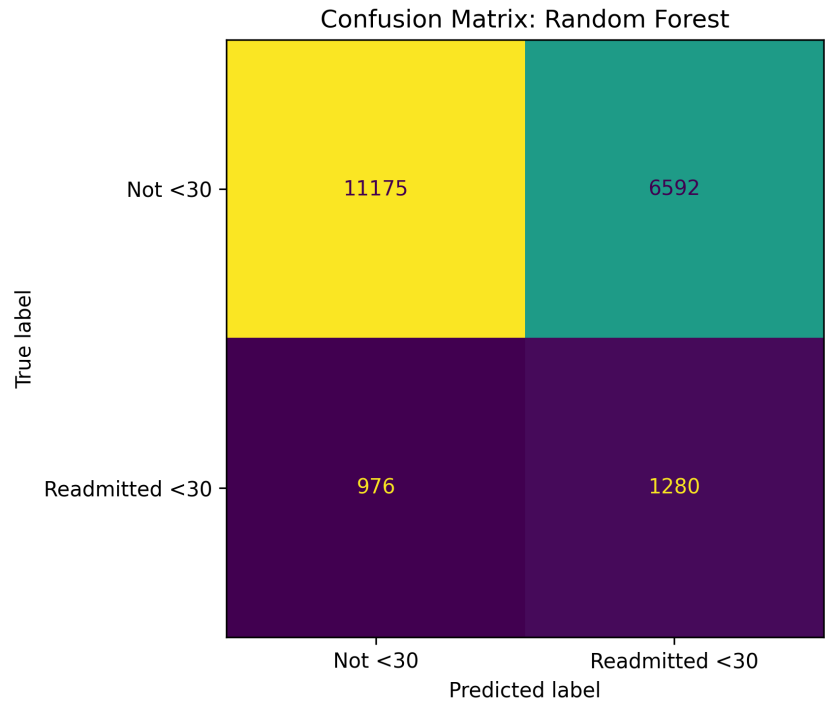


Fig. 3.
Confusion Matrix of Base Decision Tree vs Tuned Decision Tree





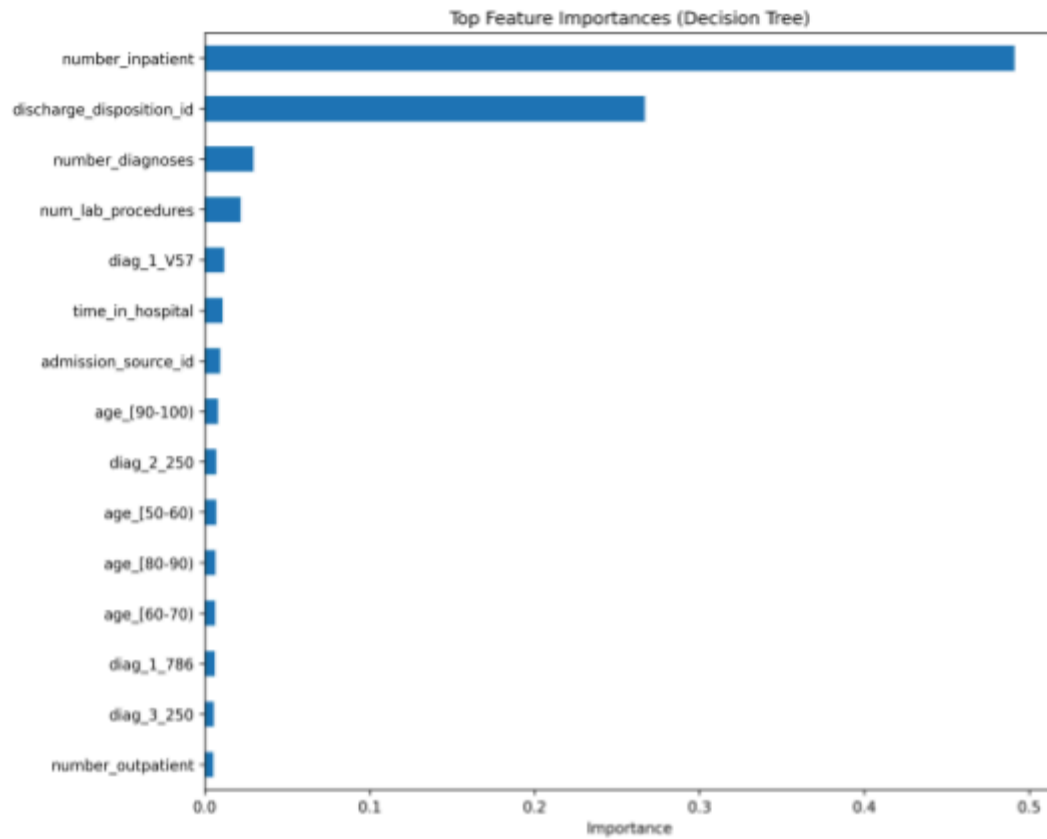


Fig. 8.
Feature Importance of Decision Tree Model

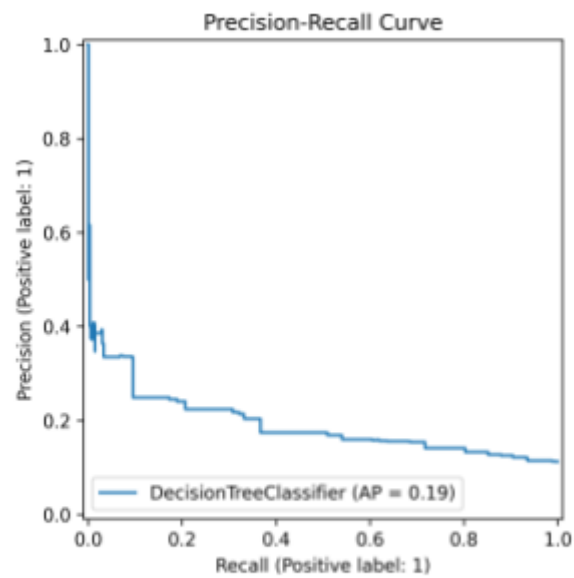


Fig. 9
Precision-Recall Curve of Decision Tree

6. Discussion

6.1. Interpretation of Results

The Decision Tree model achieved the highest accuracy, but had the lowest recall. This means that although it classified many majority class patients, it failed to identify most patients who were actually readmitted within 30 days. In essence, we found this particular metric to be misleading, since a model can achieve high accuracy by mostly predicting the majority class, despite performing poorly at identifying patients who are actually readmitted.

We decided to address this limitation of the Decision Tree model through hyperparameter tuning. By applying class weighting and limiting model complexity, the tuned model improved both recall and F1 score, allowing it to better identify readmitted patients within 30 days. While this led to a decrease in accuracy, we believe that the tradeoff is important in this healthcare setting where correct identification of high-risk patients matters more than minimizing false positives.

Analysis of feature importance for the Decision Tree model (Fig. 8) allowed better insight into factors that influenced readmission predictions. The model showed that the number of inpatient visits and discharge conditions were the most impactful factors of readmission with the number of inpatient visits having significantly higher importance than all other variables. In a healthcare context, these findings align with expectations that patients with frequent hospital visits end up more likely to be readmitted. This demonstrates the utility of the Decision Tree model in providing interpretability which is key when applying machine learning in the healthcare setting. These findings allowed us to understand why the tuned model was able to improve recall as much as it did since it focused more on clinically meaningful variables associated with the risk of being readmitted.

As a result, we believe recall and F1-scores are more useful metrics for this project. In a healthcare setting, misidentifying a patient who is likely to be readmitted could be detrimental, but a false flag is simply an inconvenience. In particular, recall indicates how correctly it identifies the largest proportion of patients who were readmitted within 30 days.

In contrast to the baseline Decision Tree model, the Gradient Boosting model achieved the highest F1-score, recall, and precision - and performed the best overall. Random Forest and Logistic Regression lagged behind, but served as useful baseline models.

A noteworthy mention is the nonlinearity between training time and performance metrics. For example, the Logistic Regression model took 5.29 seconds to train, which is nearly 22 times faster than Gradient Boosting. Despite this, Gradient Boosting's F1-score is only 8 percent higher than Logistic Regressions. In the case of a less important scenario, this tradeoff in

time-to-train and performance gains is likely inconsequential, but in this healthcare scenario, we believe the tradeoff is absolutely worth it.

7. Task Division

Kristine Alba: Led data preprocessing, including handling missing values, feature encoding, feature scaling, duplicate removal, and class imbalance handling. Implemented Logistic Regression and contributed to the methodology and preprocessing sections of the report.

James Johnson: Tuning of Gradient Boosting hyperparameters, generated visualizations of the metrics, ROC curves, and confusion matrices for all models. Wrote comparative analysis of models and discussion of results.

Josue Favela-Pacheco: Designed and implemented Random Forest model, including hyperparameter tuning using RandomizedSearchCV with cross-validation. Implemented performance evaluation metrics, accuracy, precision, recall, F1-score, and ROC-AUC, with runtime analysis.

Clinton Dela Cruz: Implemented and tuned the Decision Tree model using GridSearchCV, provided evaluation on performance of the model the specified metrics, generated and analyzed visualizations for the decision tree which included feature importance, confusion matrix, and precision-recall curve and contributed to interpretation of results and discussion of model performance

8. Conclusion

This project explored the use of machine learning models to predict hospital readmission for diabetes patients using structured healthcare data. Several preprocessing techniques were applied, including handling missing values, one-hot encoding categorical variables, feature scaling, duplicate removal, and addressing class imbalance through class weighting.

Four supervised learning models were implemented and evaluated: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. While some models initially achieved high accuracy, further analysis showed that accuracy alone was not a reliable metric due to the imbalanced nature of the dataset.

Among the evaluated models, Gradient boosting achieved the best overall predictive performance, with the highest recall, precision, and F1-score. Random Forest and Logistic Regression also performed well and demonstrated the importance of balancing computational efficiency with predictive capability. The tuned Decision Tree model also illustrated a strong

performance, achieving competitive recall and F1 score while also providing that interpretability through the feature importance and decision rules.

The results of this project emphasize the importance of selecting appropriate evaluation metrics in healthcare applications, where correctly identifying high-risk patients is more important than overall classification accuracy. Future work could explore advanced imbalance handling techniques such as SMOTE, additional feature engineering, and deep learning approaches to further improve predictive performance.

9. References

<https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008>

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